

SniffySquad: Patchiness-Aware Gas Source Localization with Multi-Robot Collaboration

YUHAN CHENG, Shenzhen International Graduate School, Tsinghua University, Shenzhen, China
XUECHENG CHEN, Shenzhen International Graduate School, Tsinghua University, Shenzhen, China
YIXUAN YANG, Electrical and Computer Engineering, Duke University, Durham, United States
HAOYANG WANG, Shenzhen International Graduate School, Tsinghua University, Shenzhen, China
JINGAO XU, The University of Hong Kong, Hong Kong, Hong Kong
CHAOPENG HONG, Tsinghua Shenzhen International Graduate School, Shenzhen, China
SUSU XU, Johns Hopkins University, Baltimore, United States
XIAO-PING ZHANG, Tsinghua Shenzhen International Graduate School, Shenzhen, China
YUNHAO LIU, Tsinghua University, Beijing, China
XINLEI CHEN, Tsinghua Shenzhen International Graduate School, Shenzhen, China

Gas source localization is pivotal for the rapid mitigation of gas leakage disasters, where mobile robots emerge as a promising solution. However, existing methods predominantly schedule robots' movements based on reactive stimuli or simplified gas plume models. These approaches typically excel in idealized, simulated environments but fall short in real-world gas environments characterized by their patchy distribution. In this work, we introduce *SniffySquad*, a multi-robot olfaction-based system designed to address the inherent patchiness in gas source localization. *SniffySquad* incorporates a patchiness-aware active sensing approach that enhances the quality of data collection and estimation. Moreover, it features an innovative collaborative role adaptation strategy to boost the efficiency of source-seeking endeavors. Extensive evaluations demonstrate that our system achieves an increase in the success rate by 20%+ and an improvement in path efficiency by 30%+, outperforming state-of-the-art gas source localization solutions.

CCS Concepts: • **Human-centered computing** → **Ubiquitous and mobile computing**; • **Computer systems organization** → **Robotic autonomy**; • **Networks** → **Sensor networks**;

Yuhan Cheng and Xuecheng Chen contributed equally to this research.

This article was supported by Yunnan Forestry and Grassland Science and Technology Innovation Joint Special Project NO. 202404CB090017, the Natural Science Foundation of China under Grant 62371269 and Tsinghua Shenzhen International Graduate School Cross-disciplinary Research and Innovation Fund Research Plan JC20220011.

Authors' Contact Information: Yuhan Cheng, Shenzhen International Graduate School, Tsinghua University, Shenzhen, China; e-mail: cyh22@mails.tsinghua.edu.cn; Xuecheng Chen, Shenzhen International Graduate School, Tsinghua University, Shenzhen, China; e-mail: chenxc24@mails.tsinghua.edu.cn; Yixuan Yang, Electrical and Computer Engineering, Duke University, Durham, North Carolina, United States; e-mail: yixuan.yang@duke.edu; Haoyang Wang, Shenzhen International Graduate School, Tsinghua University, Shenzhen, Guangdong, China; e-mail: haoyang-22@mails.tsinghua.edu.cn; Jingao Xu, The University of Hong Kong, Hong Kong, Hong Kong; e-mail: xujingao13@gmail.com; Chaopeng Hong, Tsinghua Shenzhen International Graduate School, Shenzhen, Guangdong, China; e-mail: hongcp@sz.tsinghua.edu.cn; Susu Xu, Johns Hopkins University, Baltimore, Maryland, United States; e-mail: sxu83@jhu.edu; Xiao-Ping Zhang, Tsinghua Shenzhen International Graduate School, Shenzhen, Guangdong, China; e-mail: xpzhang@ieee.org; Yunhao Liu, Tsinghua University, Beijing, Beijing, China; e-mail: yunhao@tsinghua.edu.cn; Xinlei Chen (corresponding author), Tsinghua Shenzhen International Graduate School, Shenzhen, Guangdong, China; e-mail: chen.xinlei@sz.tsinghua.edu.cn.



This work is licensed under a Creative Commons Attribution 4.0 International License.

© 2026 Copyright held by the owner/author(s).

ACM 1550-4859/2026/02-ART8

<https://doi.org/10.1145/3786599>

Additional Key Words and Phrases: Gas source localization, mobile autonomous systems

ACM Reference Format:

Yuhan Cheng, Xuecheng Chen, Yixuan Yang, Haoyang Wang, Jingao Xu, Chaopeng Hong, Susu Xu, Xiaoping Zhang, Yunhao Liu, and Xinlei Chen. 2026. SniffySquad: Patchiness-Aware Gas Source Localization with Multi-Robot Collaboration. *ACM Trans. Sensor Netw.* 22, 2, Article 8 (February 2026), 27 pages. <https://doi.org/10.1145/3786599>

1 Introduction

Rapid and accurate responses to gas leak incidents are essential for safeguarding human and environmental health [1, 2], as leaked gases can rapidly create highly flammable or toxic conditions, posing significant risks of explosions and poisoning [3, 4]. For instance, in the United States alone, 2,600 gas leakage incidents have been reported, with 328 resulting in explosions and 122 fatalities [5]. A key aspect of quick response requires localizing the gas source, which involves analyzing the concentration and distribution of the gas in the air to trace it back to its origin. With the knowledge of source locations, subsequent mitigation operations, such as shutting off valves or sealing the leaks, can be conducted more logically, efficiently, and safely.

Conventional **gas source localization (GSL)** solutions fall into two categories: (i) human expert-based solutions assign human operators to engage in affected areas. These laborious and perilous activities not only increase the risk of misinterpretation but also imperil the safety of the operators [6]; and (ii) **wireless sensor network (WSN)**-based methods utilize preinstalled static sensors to detect the gas source by monitoring gas concentration readings [7]. However, this approach is constrained by spatial resolution limitations, particularly in environments with extensive and intricate pipeline networks that present numerous potential leakage points.

In this work, we aim to devise an effective strategy for collaboratively scheduling multiple olfactory robots to localize the gas source in real-world environments. Particularly, we utilize the mobility and collaborative capability of multiple robots to gather more information efficiently, enhancing the synergy between their sensing and scheduling abilities.

We take gas leakage in a factory with intricate pipelines as an example scenario, as shown in Figure 1. Here, a fleet of robots equipped with gas concentration sensors and wind sensors are dispatched to search the surroundings and gather gas sensory data. By analyzing and interpreting the environmental conditions, the robots autonomously navigate to pinpoint the source of the gas leakage. However, translating this idea into a practical system is non-trivial and faces two challenges:

— **The patchy nature of gas plumes confuses and traps the robots.** The gas landscape is characterized by a *patchy structure* [8, 9], as evidenced by field measurements of gas concentration (see Figure 2). As seen, gas plumes fragment into disjointed patches and form distinct and scattered areas. From the perspective of source-seeking robots, the concentration measured along the horizontal centerline exhibits pronounced fluctuations as the distance to the source increases, since gas patches are separated by regions where gas concentration is below detectable levels. This intermittent characteristic introduces uncertainty in determining the source direction when robots devise paths using noisy and local sensory data, ultimately leading to their entrapment within these patches and failure to complete GSL. Existing solutions, including bio-inspired [10, 11] and probabilistic model-based approaches [12, 13], either rely on local gas concentration gradients or oversimplified gas dispersion models, limiting their effectiveness to idealized scenarios with continuous gas concentration fields.

— **Trade-off between source localization effectiveness and search efficiency for collaborative robots.** Since gas distribution varies continuously over space and time, continuous spatiotemporal measurement is infeasible. Therefore, robots must strike a balance between two



Fig. 1. Mobile olfactory robots autonomously search for and navigate towards the source of gas leakage.

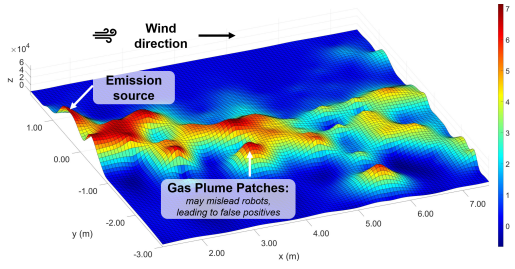


Fig. 2. **Spatial characteristics of gas concentration.** We conducted a proof-of-concept experiment to check gas characteristics in our indoor testbed, with a gas emission device at $(x, y) = (1.0, 0.0)$ and winds blowing along the x -axis. The heatmap illustrates concentration values, representing the number of particles with a diameter $>0.3\mu\text{m}$ in 0.1L of air.

conflicting objectives within limited sampling: (i) identifying new potential gas source positions and (ii) excluding false positive source positions. Achieving these goals simultaneously complicates the localization of the true gas source efficiently and effectively. Specifically, the former goal necessitates extensive exploration and sampling of gas data, while the latter goal requires utilizing previously acquired information to reach and verify the estimated source position's probability. Current multi-robot GSL solutions tend to exhibit either clustered or dispersed collaborative behaviors, thereby undermining the simultaneous achievement of both objectives [11, 14, 15].

Remark. As far as we are aware, previous works develop operating principles that neglect the patchy nature of gas plumes, based on which they schedule agents without fully harnessing their versatility and collaborative capabilities at a system level.

To tackle the above challenge, we design and implement *SniffySquad*, a collaborative olfactory robot scheduling system for GSL. Benefiting from *SniffySquad*, a team of olfactory robots can adapt to field patchiness and dynamically adjust their movements accordingly, enabling efficient and effective emission source localization. In general, *SniffySquad* excels in the following two aspects.

— **At the individual level**, we propose a *Patchiness-aware active sensing* method, which refines the probabilistic source estimation by incorporating the patchy characteristic of gas in each robot's moving strategy. Inspired by the principles of Langevin MCMC [16], we adjust robots' gradient-based moving direction with a regulation term, allowing them to escape false positive source positions and thereby collect more informative sensing data.

— **At the team level**, we design a *Potential-instructed collaborative roles adaptation* strategy that further enhances the source-seeking efficiency by adjusting robots' roles based on their spatial distribution and measurements in the past. Each robot in the team adopts either a fine-grained search role to inspect spurious signals (i.e., exploiter) or a coarse-grained search role to discover potential new sources (i.e., explorer). These roles are adaptively adjusted based on the estimated probabilities of each robot's proximity to the emission source, thereby parallelizing the exploration of the environment and exploitation of collected information in a flexible manner.

We evaluate the performance of *SniffySquad* and compare it with **state-of-the-art (SOTA)** baseline methods through experiments on a real-time multi-robot testbed (12 hours) and extensive physical feature-based simulations (750 runs). Experiment results show that our system outperforms all baselines, achieving a 20%+ **success rate (SR)** improvement and improving **path efficiency (PE)** by $> 30\%$.

We summarize the contributions of this article as follows:

- We propose *SniffySquad*, a collaborative multi-robot olfactory sensing system for accurate and efficient GSL, specifically designed to handle the patchy characteristics in real-world gas fields.
- We introduce a patchiness-aware active sensing method that refines probabilistic source estimation by incorporating gas patchiness in each robot’s moving strategy. Building on this design, we further devise a potential-instructed collaborative roles adaptation strategy, which parallelizes the exploration of the environment and exploitation of collected information in an adaptive manner.
- We develop a prototype system and evaluate *SniffySquad* through a real-world testbed and a physical-feature-based gas dispersion simulator. Extensive evaluation results show its effectiveness and superior performance.

The remainder of the article is organized as follows. We first introduce the background, motivation, and related works in Section 2. In Section 3, we present preliminaries, followed by the system overview in Section 4. In Sections 5, 6, we introduce the algorithm design, involving detailed descriptions of the patchiness-aware active sensing and potential-instructed collaborative roles adaptation. Section 7 showcases the implementation and evaluation. Finally, Section 9 concludes the article.

2 Background and Related Work

2.1 Characteristics of Gas Concentration Distribution

Spatial characteristics of gas concentration in real-world are depicted in Figure 2. The data are measured in our testbed, which will be introduced in Section 7.1. Figure 2 shows that gas distribution exhibits patchiness, which is caused by turbulent atmosphere stretching and fragmenting plumes into disjointed patches spatially [8]. This characteristic may mislead source-seeking robots into falsely identifying them as the actual gas emission source.

Note that characteristics of gas concentration distribution described by models commonly used in previous GSL works differ significantly from those observed in practical environments. We first introduce the model that existing work relies on. To model the concentration of a substance flowing in a fluid, the transportation process is typically adopted, which is described by the partial differential convection-diffusion equation $\frac{\partial c}{\partial t} = \nabla \cdot \mathbf{u}c - \nabla \cdot \Gamma \nabla c + S$. Here scalar c is the gas concentration, $\mathbf{u}$ is the velocity vector (u, v, w), ∇ and $\nabla \cdot$ represents the gradient and the divergence of the c , and S is the source of a scalar. While this equation can be employed for numerical simulation in **computational fluid dynamics (CFD)**, as highlighted in the preceding section, probabilistic GSL approaches necessitate an analytical model rather than a transportation process for numerical simulation. To this end, previous works [14, 17] solve a simplified transportation equation by assuming steady-state conditions and assuming no degradation, yielding the Gaussian plume model [18].

However, this oversimplified model can only be used to model gas concentration landscapes at microscopic scales, where molecular diffusion generates smooth changes in concentrations of chemicals [10]. In contrast, as shown in Figure 2, at distances on the order of meters or larger, gas plumes are significantly dispersed, resulting in patchy and discontinuous plumes at macroscopic scales.

2.2 Mobile Robot Olfaction-based Gas Source Localization

Current approaches can be categorized into two classes, which will be introduced first, reactive bio-inspired and probabilistic algorithms. Then, multi-robot GSL methods are summarized, introducing how collaboration is incorporated [19, 20].

2.2.1 Reactive bio-inspired GSL. Bio-inspired algorithms draw inspiration from the behavior of insects or rodents. They involve chemotaxis, anemotaxis, or combining both [10]. For chemotaxis, such strategies have the robot turn to the opposite direction if it experiences a declining odor concentration gradient along its trajectory. Chemotaxis fails when odor trails are broken, olfactory stimuli are sporadic, and consequently, gradients are absent. Anemotaxis-driven navigation can be divided into surge (the robot moves upwind), casting, spiral, and zigzagging [11]. For casting, a robot in the plume moves upwind until it loses the plume, then it turns and moves cross-wind until it hits odor packets again. When a robot loses the plume, it moves along a spiral or zig-zag to reacquire the plume. Bio-inspired methods are reactive in nature, making them susceptible to intermittent gas concentration signals caused by spurious gas patches.

2.2.2 Probabilistic gas source localization methods. According to models used for the environment's representation, probabilistic GSL methods can be categorized into parametric and non-parametric algorithms [12].

— **Parametric algorithms**, which rely on an **atmospheric transport and diffusion (ATD)** model [21, 22] mapping each point in the environment to the expected gas concentration there, assume a finite set of parameters. In the context of GSL, the set of parameters is called source term, involving a combination of the position of the source, the release rate, the down-wind direction and velocity, and 3-dimensional diffusion terms in different directions (x, y, and z). Prevalent plume models include the isotropic plume model [23] and the Gaussian plume model [24].

— **Non-parametric algorithms.** They model the spatial distribution of the gas plume rather than focus on terms parameterizing the gas source, thus can not be defined in terms of a finite set of parameters. These algorithms originate from the problem of **gas distribution mapping (GDM)**. Popular algorithms include Kernel DM+V [25] (and its variants) and **Gaussian Markov random fields (GMRF)** based spatial modeling [13, 26].

Parametric methods are mostly computationally efficient and exhibit simple analytical forms. However, this oversimplified plume model fails to account for the stochasticity of gas emissions and the spatial nonuniformity of wind fields. It renders this type of algorithms idealistic. Moreover, these methods fail under the existence of multiple sources or the inclusion of walls and obstacles. The latter fact restrains the methods' applicability to indoor places, such as factories. In contrast, non-parametric algorithms make less assumptions about the atmospheric environment and the gas source characteristics, hence are much more flexible to be applied in various scenarios. However, without much prior knowledge embedded in non-parametric algorithms, inference at areas that are previously unvisited are typically inaccurate. Moreover, the extensive parameter space poses computational challenges.

According to planning strategies used to seek the source, probabilistic GSL methods can be categorized into gradient-based and information-based algorithms.

— **Gradient-based probabilistic GSL planners** resemble the idea of chemotaxis but differ from it in the way they compute gradients. In chemotaxis, robots utilize merely measurements along its trajectory to identify whether the concentration is rising or declining. In probabilistic GSL algorithms, robots can leverage the probability model to calculate the direction, either analytically or numerically. For example in [27], particle filter **biased random walk (BRW)** modifies BRW [28] to utilize an analytical gradient computed from source term estimation results based on the Gaussian plume model.

— **Information-based GSL planners** select actions to maximize the expected information gain (i.e., the uncertainty reduction). The algorithm can be implemented by optimizing **upper confidence bound (UCB)** [29], entropy such as mutual information [30], or Fisher information [31].

Table 1. Comparison of GSL Approaches

GSL approach	Representative work	Weakness in handling patchiness
Bio-inspired	Surge-Cast [10, 34]	Heuristic and not tailed for GSL in patchy plumes
Information gain-based	Infotaxis [9, 35]	Entropy can be unmeaningful when information is unreliable
Population-based	PSO for GSL [15, 36]	Robots tend to move towards the same local optimum

2.3 Cooperative Gas Source Localization methods

An apparent way of increasing GSL efficiency and accuracy is by utilizing multiple robots as a team [32, 33]. Some directly extend single robot GSL algorithms by spanning the action space. Other methods, tailored for cooperative GSL, are listed here. (i) **Particle Swarm Optimization (PSO)**-based algorithms [15] are inspired by the bird flock searching for food. If an agent discovers a better pattern (potential food source), others adjust their movements to follow it. When robot teams adopt this strategy for GSL, they will have a tendency to move towards the same local optimum and are considered to lack cooperation [11]. (ii) Another type of approach [14] achieves cooperation by dispatching robots to investigate several likely source positions separately, so that they can either be ruled out or considered more thoroughly.

In summary, these multi-robot GSL methods induce either gathering or scattering behavior, so they lack efficiency in achieving both goals. Technically, each individual in the team acts similarly and are not assigned heterogeneous roles. In contrast, our work empowers team members to adopt different but complementary roles, enhancing individual diversity and overall synergy. This distinction highlights the advantage of a more strategic and versatile approach to team coordination.

We summarize existing GSL methods and their limitations in handling plume patchiness in Table 1.

3 Preliminaries

3.1 Definitions

We denote gas source position and robot's position at time t as x_s and $x^{(i)}(t)$, respectively. The total time spent for GSL is represented by final timestep t_f . The trajectory of a robot and its corresponding sensory measurements are denoted by $\mathbf{z}_{1:t}$ and $\mathbf{x}_{1:t}$.

We express measurements at position x as $z(x)$. We denote measurement data as $z = (z^c, z^w)$, where gas concentration and airflow direction are denoted by c and $\mathbf{u} = (u, v, w)$, respectively. The system is only accessible to a belief $b(t)$ incorporating sensory measurements $\mathbf{z}_{1:t}$ along past trajectories $\mathbf{x}_{1:t}$ due to partial observability. It follows the Bayesian estimation framework by continually updating the likelihood of each position being situated within the gas plume region based on incoming measurements z gathered from sensors equipped on the mobile robots. Mathematically, $p(x|z_{1:t}) \propto p(x|z_{1:t-1})p(z_t|x, z_{1:t-1})$.

Important symbols used in the article are listed in Table 2.

3.2 Problem Formulation

The goal of *SniffySquad* is to minimize the search time, defined as the duration from departure to a robot reaching the vicinity of a gas source. A search trial terminates when a robot's position is within a predefined threshold of the gas source or the predefined time limit is reached. Mathematically, a successful search indicates that $\exists i \in \{1, 2, \dots, M\}$ s.t. $\|x^{(i)}(t_f) - x_s\| \leq d_\epsilon$, where d_ϵ is the distance

Table 2. List of Important Notations Used in the Article

Notation	Description
M	Total number of robots
$x(t) \in \mathcal{X}$	Position coordinates at time t
$\mathbf{x}$	Positions of all M robots $= [x^{(1)}, x^{(2)}, \dots, x^{(M)}]^T$
x_s	Gas source position
c	Gas concentration
$\mathbf{u}$	Wind velocity vector $= (u, v, w)$
$z \in \mathcal{Z}$	Sensory measurement collected from mobile robot nodes (consisting of gas concentration z^c and the wind vector $z^w = (z^c, z^w)$)
t_f, t_0	Time of task termination and departure
$U(x)$	Neighborhood of position x
$p(x)$	Probability of the gas source being located at x
$\Phi : \mathcal{X} \rightarrow \mathbb{R}$	Potential function
$\tau, \boldsymbol{\tau}$	Temperature parameter in Langevin diffusion; Temperatures of all M robots $= [\tau^{(1)}, \tau^{(2)}, \dots, \tau^{(M)}]^T$
$a, s(i, j)$	Swapping rate (i.e., role-exchange probability) between robot i and robot j
μ, σ	Deterministic and stochastic terms in the generic stochastic differential equation (SDE)
dB, ξ	Brownian motion in continuous and discrete time space
η, h	Temporal and spatial discretization granularity
$\pi(\cdot)$	Invariant distribution of MCMC sampling

threshold. Therefore, the system objective can be formulated as:

$$\begin{aligned}
& \min_{\mathbf{x}(t), \mathbf{u}(t)} \quad t_f \\
& \text{s.t.} \quad b(t) = \text{bel}(\mathbf{z}_{1:t}, \mathbf{x}_{1:t}), \\
& \quad u^{(i)}(t) = g(x^{(i)}(t), x^{(-i)}(t), b(t)), \\
& \quad \dot{x}^{(i)} = f(x^{(i)}, u^{(i)}), \\
& \quad \mathbf{x}(t_0) = \mathbf{x}_0, \mathbf{x}(t_f) = \mathbf{x}_f, \\
& \quad x \in \mathcal{X}, u \in \mathcal{U}, t_f > 0,
\end{aligned} \tag{1}$$

The global gas source estimator is represented as $\text{bel}(\cdot)$. The collaborative planner is denoted as $g(\cdot)$, which determines the movements of robot i , $u^{(i)}(t)$, according to the estimation results $b(t)$, its own position $x^{(i)}(t)$, and other robots' positions $x^{(-i)}(t)$. The control logic is denoted as $f(\cdot)$, describing the dynamics of specific robots.

4 System Overview

SniffySquad is designed to schedule a team of autonomous mobile robots equipped with gas detection sensors to efficiently navigate toward and localize gas leakage sources. The system utilizes a layered approach for functionality and resource management: (i) *agent layer*, where each mobile agent self-localizes and determines its next movement in real-time; (ii) *edge layer*, which communicates with the agents, fuses local data, and optimizes global gas source estimation to guide navigation. This structure ensures effective environmental information collection and processing to achieve the gas localization objective.

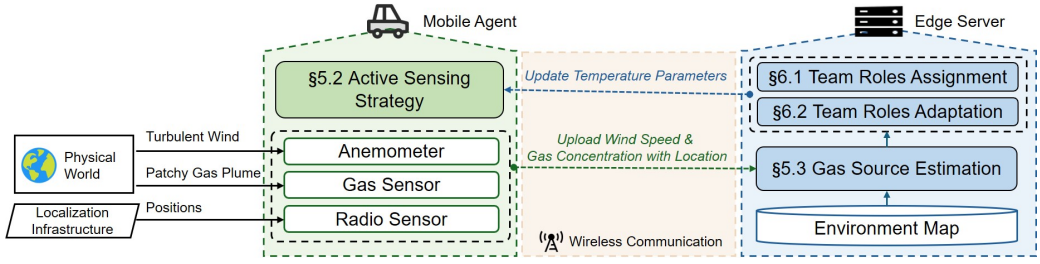


Fig. 3. System overview.

Architecture. Figure 3 shows the system architecture. *SniffySquad* provides system support on both the mobile agent and edge server side, as described below.

—On the mobile agent side, each robot serves as a flexible, mobile olfactory system for navigating toward potential gas leak regions. The hardware configuration includes a gas sensor and an anemometer for measuring gas concentration and estimating airflow speed and direction. Each robot’s real-time position is obtained through an **Ultra-Wide Band (UWB)** tag, supported by pre-deployed UWB anchors for localization. The robot runs the active sensing strategy module, which receives high-level role-specific parameters from the edge server. These parameters guide the robot’s navigation strategy, enabling the robot to adjust its movement and generate low-level control commands based on the received information.

—On the edge side, the edge server coordinates the robot team by executing modules for gas source estimation, team role assignment, and role adaptation. It maintains a global probability map, which associates each grid in the environment with the likelihood of containing the gas source. The server fuses local sensor data (gas, wind, and location) and continuously updates the probability of each position within the gas plume based on incoming measurements. Using this updated estimation, the robot adaptation module manages the coordination of robot behaviors and dispatches motion plans to the mobile robots, ensuring alignment with the current global gas source assessment.

Workflow. Robots initially gather sensory data, including gas concentration, wind speed, and direction. The probabilistic gas source estimation module updates source estimates using the filtered measurements. The team roles adaptation module reconfigures robot roles when their potential for source localization misaligns with their assigned roles, updating each robot’s temperature parameters accordingly. Finally, the robots generate new sensing directions through the active sensing strategy module based on the updated temperature parameters.

5 Patchiness-aware Active Sensing

To tackle the patchy nature of gas plumes, we design a patchiness-aware active sensing method for individual robots to enable resilient GSL. The strategy is inspired by **Markov chain Monte Carlo (MCMC)** sampling based nonconvex learning. In this section, we first recast the GSL problem to MCMC sampling (Section 5.1), then we describe the algorithm design, that is, tailored for olfactory mobile robots sensing actively in realistic plumes (Section 5.2). At last, we introduce the probabilistic gas source estimation method, which constitutes sampling’s target distribution (Section 5.3). Figure 4 illustrates the workflow of the latter two sections.

5.1 Recast GSL to MCMC Sampling

In this part, we first introduce a surrogate optimization via which the problem formulated in Equation (1) can be solved. Regarding the patchy nature of gas plumes, the surrogate optimization

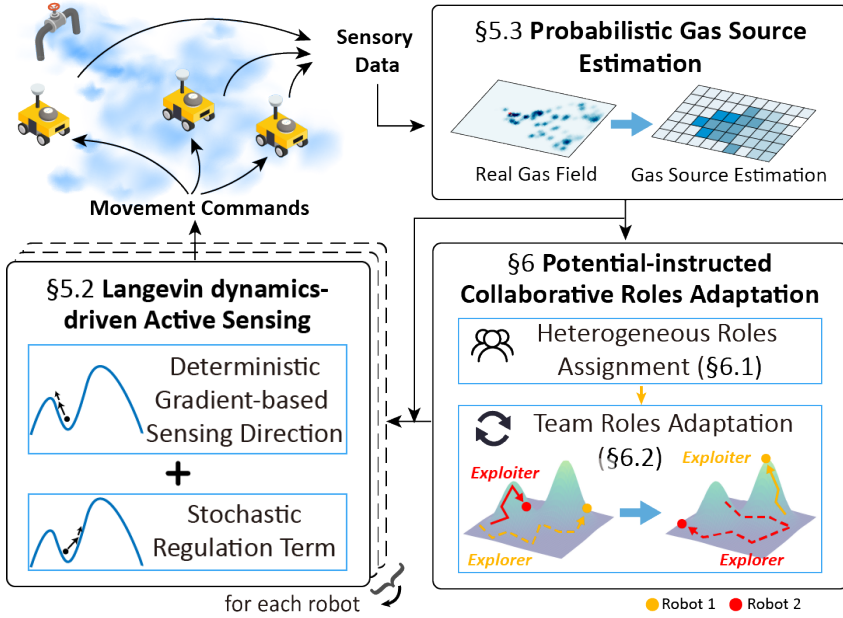


Fig. 4. Illustration of the algorithm.

is then reframed as MCMC sampling. At last, we describe the Langevin MCMC, based on which we design our algorithm tailored for mobile olfactory robots.

5.1.1 Surrogate optimization. Our task, GSL, involves identifying a position with the highest probability of locating the source of a gas emission. From this perspective, the objective of Equation (1) can be realized by optimizing the likelihood of being the source location, with respect to position x . An efficient optimizing procedure leads to an efficient source seeking algorithm.

Specifically, instead of minimizing the terminating time t_f directly, we can optimize a surrogate function $\Phi(x) : \mathcal{X} \rightarrow \mathbb{R}$, which assesses the likelihood of position x being the source location (position x with a small value of $\Phi(x)$ has a high probability of being the source of gas leakage). Thus, the core task can be reinterpreted, leading to the formulation:

$$\min_{x \in \mathcal{X}} \Phi(x). \quad (2)$$

We describe the probabilistic estimator and the design of potential $\Phi(x)$ in Section 5.3.

5.1.2 Optimization via sampling. In the context of GSL, however, the optimization task presents notable challenges attributable to two factors. First, the potential function exhibits multimodal characteristics and is far from convex due to the patchy gas plume. Second, the potential function available is noisy and stochastic [37]. Specifically, the robots rely on measurements obtained from onboard sensors, which are susceptible to noise and provide partial observations. Consequently, conventional algorithms suitable for convex objectives and methods taking deterministic movements can be trapped and/or misled.

To deal with the challenges posed by the nonconvex and stochastic nature of the potential function, reframing the optimization problem to a MCMC sampling problem [38] is one promising approach. Monte Carlo sampling has been shown to be a powerful method for nonconvex and multimodal optimization [16, 39, 40], as well as noises and randomness in the potential function

[37]. Specifically, sampling can be faster than optimization for some nonconvex objectives [39]. Moreover, for noisy objectives whose randomness might arise from inherent function noise, MCMC sampling algorithms can be efficient by injecting the right amount of noise into each iterative update step, providing a means of coping with inherent noise. For instance, stochastic **gradient descent (GD)** methods [37] are designed to cope with such randomness.

A key step in MCMC sampling is constructing a Markov chain that has the desired probability distribution $\pi(x)$ as its equilibrium distribution. Under this condition, one can draw samples from the intended distribution by recording states from the chain. Specifically, convergence to equilibrium means that the Markov chain “forgets” about its initial state and the sequence of states after convergence to distribution $\pi(x)$.

5.1.3 Langevin diffusion. To translate MCMC sampling methods to robot movement algorithms, we design the strategy based on Langevin diffusion MCMC. Now, we first introduce the generic diffusion MCMC, then introduce the Langevin diffusion.

Diffusion MCMC sampling methods simulates a Markov chain using diffusion processes, a concept from statistical physics describing the movement of particles. A diffusion process is defined by a SDE, whose general form can be expressed as:

$$dx(t) = \mu(x(t))dt + \sigma(x(t))dB(t), \quad (3)$$

where $x(t)$ is the positions of the particle. This SDE consists of a deterministic drift term $\mu(x(t))$ and a stochastic term, which is proportional to the standard Brownian motion $dB(t)$ and scaled by a factor $\sigma(x(t))$. The standard Brownian motion $\{B(t)\}_{t \geq 0}$ is a stochastic process with properties (i) movement during any time interval (s, t) ($s < t$), $B(t - s) \doteq B(t) - B(s)$ is a normal random variable, i.e., $B(t - s) \sim \mathcal{N}(0, t - s)$, and (ii) the increments of different time intervals are independent.

The Langevin dynamic, with respect to the potential function $\Phi(x)$, is a diffusion process exhibiting equilibrium behaviour. It is given by a SDE:

$$dx(t) = -\nabla\Phi(x(t))dt + \sqrt{2\tau}dB(t), \quad (4)$$

where $\nabla\Phi(x(t))$ is the gradient of the potential function at position $x(t)$, $\{B(t)\}_{t \geq 0}$ is a standard Brownian motion and $\tau > 0$ is a parameter called temperature in statistical physics and thermodynamics. Notably, Equation (4) is the stochastic version of GD, differing in the incorporation of noise term, which is Gaussian with zero mean and a covariance determined by the temporal duration of the movement.

This random process $\{x(t)\}_{t \geq 0}$ defined in Equation (4) has a unique invariant distribution (a.k.a. the Boltzmann-Gibbs distribution) with density

$$\pi_\tau(x) = \frac{e^{-\Phi(x)/\tau}}{\int_{\mathcal{X}} e^{-\Phi(x)/\tau} dx} = \frac{1}{Q} e^{-\Phi(x)/\tau}. \quad (5)$$

Here the denominator Q is the normalization constant. With any initial distribution of $x(t)$, the limiting distribution of X_t converges to $\pi_\tau(x)$ by running the diffusion process. In other words, our solution is always expected to fall into the neighborhood of a global minimum of $\Phi(x)$ with high probability.

5.2 Langevin Dynamics-driven Active Sensing Strategy

In this part, we describe the novel active sensing strategy for GSL with resiliency to patchy gas plume caused by atmospheric turbulence and noisy partial observations from robots.

As illuminated in the previous section, gas source search and localization can be recast to drawing samples from $\pi(x) = e^{-k\Phi(x)}$, so that random samples drawn from $\pi(x)$ are attracted to the peak in the distribution $\pi(x)$, i.e., the global minimum of $\Phi(x)$. This relationship is exemplified when

$\pi(x) = \delta(x^*)$, where $x^* = \arg \min \Phi(x)$, as $k \rightarrow +\infty$. In particular, the Dirac delta equilibrium distribution concentrates at x^* (the global minimum of $\Phi(x)$ and hence the position with the maximum likelihood of locating the source) and the chain “forgets” about initial position of the mobile robot.

We aim to develop an active sensing strategy for source-seeking robots driven by Langevin dynamics. However, there is a gap between an algorithm tailored for robots and MCMC sampling due to the difference between particle dynamics and the mobile robots’ motion characteristics. Langevin diffusion is developed to describe the motion of particles (or the update of parameters of machine learning models) by defining a continuous-time stochastic process, thus Langevin MCMC can not be directly applied. To establish strategy suitable for robots determining their sensing direction, we first discretize the diffusion process and subsequently generate a sequence of positions for mobile robots to follow.

In the first step, we attain a discrete-time algorithm through the Finite Difference Method [41]. In detail, we first discretize the states over time from $\{x(t)\}_{t \in \mathbb{R}^+}$ to $\{x_k\}_{k \in \mathbb{Z}^+}$, where $x_k = x(k \cdot \delta t)$, and δt represents the discretization timestep size. Then, we apply the explicit, forward-time Euler scheme for temporal discretization, leading to the following formulation of the discrete SDE:

$$x_{k+1} = x_k - \eta \cdot \nabla \Phi(x_k) + \sqrt{2\eta\tau} \cdot \xi_k. \quad (6)$$

Here the stepsize $\eta > 0$ is proportional to the temporal discretization granularity δt , τ is the temperature parameter, and $\{\xi_k\}_{k \in \mathbb{Z}^+}$ signifies a sequence of standard normal random vectors. Equation (6) describes how the robot’s sensing direction evolves overtime while incorporating randomness.

In the above equation, it requires calculating the gradient of the potential function at position x to generate a sequence of trajectory points. However, the function we designed in Section 5.3 is discrete and not differentiable. To address this, we adopt the central-difference approximation for spatial discretization:

$$\nabla \Phi(x) \approx \left[\frac{\Phi(x+h, y) - \Phi(x-h, y)}{2h}, \quad \frac{\Phi(x, y+h) - \Phi(x, y-h)}{2h} \right]^T.$$

Additionally, following gradients might force the robot jump between distant positions when the slope of the potential function at current position is substantial. For robots’ path scheduling, we constrain the drift term to be below the robot’s maximal velocity.

5.3 Probabilistic Gas Source Estimation

With the reframing of the GSL task as an MCMC sampling problem, a definition of the potential function $\Phi(x)$, as stated in Equation (2) and in equilibrium distribution Equation (5), is required. For every position $x \in \mathcal{X}$ in the environment, we estimate the probability that x is located within the gas plume region. This probability estimation is denoted as $p(x)$, and subsequently, we define the potential function as: $\Phi(x) \doteq -\log(p(x))$.

To obtain $p(x)$ we employ a Bayesian estimation methodology, placing our method within the class of probabilistic GSL algorithms. Specifically, this estimator continually updates the likelihood of each position being situated within the gas plume region based on incoming measurements z gathered from sensors equipped on the mobile robots. Mathematically, $p(x|z_{1:t}) \propto p(x|z_{1:t-1})p(z_t|x, z_{1:t-1})$. In our scenario, the measurements $z = (z^c, z^w) \in \mathcal{Z} = \mathbb{R} \times \mathbb{R}^d$ collected from the mobile robots consists of the gas concentration z^c and the wind vector z^w , acquired by chemical sensor and wind sensors (anemometers), respectively. To accommodate environments of arbitrary geometries, the area is discretized into a grid of equally-sized cells.

The update procedure upon receiving new measurements is implemented based on [35], involving two sequential steps. (1) *Local estimation generation*. The first step follows two intuitive and empirical

ALGORITHM 1: *SniffySquad*.

Input: Robots' initial positions $x^{(0)}$
Output: d_ϵ -approximate gas emission source position

```

1 while  $\forall i \in \{1, 2, \dots, M\}, \|x^{(i)}(t_f) - x_s\| \geq d_\epsilon$  do
2   Collect environmental sensory measurements  $z = (z^c, z^w)$ ;
3   Update belief  $b(x)$ , and potential  $\Phi(x)$  accordingly;
4   CollaborativeRolesAdaptation();
5   for  $i$  in  $\{1, 2, \dots, M\}$  do
6     Assign a role  $\tau^{(i)}$  to robot  $i$ ;
7     Move robot  $i$  following  $x_{k+1}^{(i)} = x_k^{(i)} - \eta \cdot \nabla \Phi(x_k^{(i)}) + \sqrt{2\eta\tau^{(i)}} \cdot \xi_k$ ;
8     if  $d(x_{k+1}^{(i)}, x_s) < d_\epsilon$  then return;
9   end
10 end
11 Function CollaborativeRolesAdaptation()
12   for  $i, j$  in  $\{1, 2, \dots, M\}$  do
13     Calculate swapping rate  $s(i, j) = a \cdot \exp(\min(0, (\frac{1}{\tau^{(i)}} - \frac{1}{\tau^{(j)}}) \cdot (\Phi(x^{(i)}) - \Phi(x^{(j)})))$ ;
14     if  $\text{random}() \leq s(i, j)$  then
15       Robot  $i$  and  $j$  swap their roles (i.e., temperatures  $\tau^{(i)}$  and  $\tau^{(j)}$ );
16     end
17   end

```

rules: (a) if the robot detects gas, the likelihood of neighboring grids within the upwind direction to localize within the gas plume is increased; (b) If gas concentration is negligible or falls below a predefined threshold, grids directing to the position with previously recorded nonzero gas readings are attributed a higher likelihood of hosting gas. (2) *Global estimation propagation*. The first step generates estimations solely for $x \in U(x(t))$, where $U(x(t))$ denotes the neighborhood region around the robot's current position $x(t)$. For the remaining areas $\{x \in \mathcal{X} \setminus U(x(t))\}$, their probabilities are updated in the second step by propagating information from $U(x(t))$ to the entire environment $\mathcal{X}$ based on [35], conforming to the geometry of the area.

6 Potential-instructed Collaborative Roles Adaptation

In this part, we present how to coordinate a team of collaborative robots to enhance source-seeking efficiency, utilizing both external environmental cues and internal team dynamics. This strategy consists of two tightly-coupled modules: *Heterogeneous Roles Assignment* and *Team Roles Adaptation*. We present our algorithm in Algorithm 1 and its convergence analysis in Appendix A.

6.1 Heterogeneous Roles Assignment

The scattered, patchy nature of gas plumes and the multimodal nature of concentration distribution map require a GSL system with two capabilities, (i) discovering new potential source positions and (ii) distinguishing existing candidate positions. This gives rise to a trade-off between "global exploration" and "local exploitation" of the environment. To attain the advantages of both behaviors, we assign robots in the team with heterogeneous roles, named *explorer* and *exploiter*, respectively. Robots acting as the former role are mainly responsible for objective (i) while the latter focus on objective (ii).

Considering this, we assign different roles to robots in a team. The role assignment can be implemented by regulating the temperature parameter τ in Equation (4). Specifically, temperature τ in the motion dynamic plays a crucial part in regulating the behaviour of the robot. Robots assigned

a low temperature carry out exploitative actions while those with high temperatures engage in exploratory actions. Mathematically, for a team with M robots, consider M Markov chains driven by Langevin diffusions:

$$dx^{(i)}(t) = -\nabla\Phi(x^{(i)}(t))dt + \sqrt{2\tau^{(i)}(t)}dB^{(i)}(t), \quad (7)$$

where $i \in \{1, 2, \dots, M\}$. The role of robot i is specified by its temperature $\tau^{(i)}(t)$, which governs the extent of exploring potential gas source positions and exploiting local information to distinguish spurious gas patches. When a robot is assigned a low temperature, its motion is governed by the gradient of the potential function, the robot tends to exploit the local information of the surrounding potential field. For a robot with a higher temperature, more randomness is injected into the motion, causing it to explore the environment more actively.

The role division module, following Equation (7), have a guarantee that robots are expected to concentrate around the global minima of the potential function $\Phi(x)$. Specifically, we denote the positions of M robots as $\mathbf{x} = [x^{(1)}, x^{(2)}, \dots, x^{(M)}]^T$, and their corresponding temperature parameters as $\boldsymbol{\tau} = [\tau^{(1)}, \tau^{(2)}, \dots, \tau^{(M)}]^T$. The invariant joint distribution [40] takes the form:

$$\pi(\mathbf{x}) \propto \exp\left(-\sum_{i=1}^M \frac{\Phi(x^{(i)})}{\tau^{(i)}}\right). \quad (8)$$

The marginal stationary distribution of the low temperature robot will be attracted to the peak of the potential function $\Phi(x)$.

6.2 Team Roles Adaptation

This submodule adaptively adjusts team roles through role swapping, as illustrated in Figure 4, based on the principles of replica exchange.

Optimizing the trade-off between localization effectiveness and search efficiency requires robots in the team to adjust their roles adaptively. One way to view this is through the lens of MCMC sampling. In Equation (4), a high temperature fosters broader exploration across the global domain at the expense of reduced concentration around the minima, and vice versa. Therefore, the potential of using a fixed temperature is inherently limited. Alternatively, consider the gas source localization scenario. At an individual level, robots should exploit parsimoniously to avoid trapping in spurious patches and explore appropriately to escape from areas improbable to locate the source. From a team perspective, there should be some robots exploiting while others exploring.

In order to coordinate the team behavior systematically, we propose a role-swapping mechanism for team roles adaptation, incorporating replica exchange MCMC sampling [42]. When an exploiter i locates a position, that is, less likely to correspond to the desired gas source compared to an explorer j , this pair of robots swaps their roles. Consequently, robot i transitions into an explorer so that it can turn to more promising areas and save time, while robot j transforms into an exploiter who slows down and focuses to avoid missing the source of interest.

Formally, this is achieved by swapping the temperatures between robots exhibiting different potential. Specifically, we consider a pair of random processes, denoted as robot i and j , from the set of M Langevin diffusions. The two robots swap temperatures with probability:

$$s(i, j) = a \cdot \exp\left(\min\left(0, \left(\frac{1}{\tau^{(i)}} - \frac{1}{\tau^{(j)}}\right) \cdot (\Phi(x^{(i)}) - \Phi(x^{(j)}))\right)\right). \quad (9)$$

Here the constant $a \geq 0$ represents the swapping intensity. In this manner, robots occupying positions associated with smaller potential function values are inclined to be assigned a role of exploiter, specified by a lower temperature, and vice versa. To elucidate the exchange rule, let us

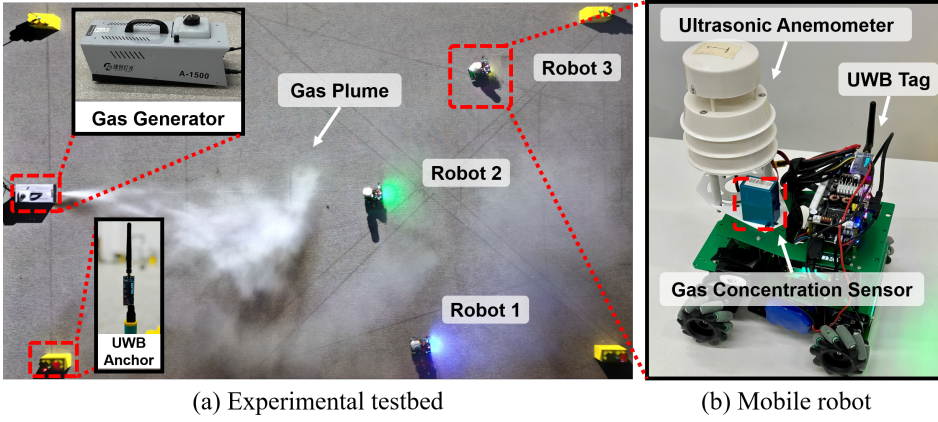


Fig. 5. Experimental testbed of *SniffySquad*.

assume, without loss of generality, that $0 < \tau^{(i)} < \tau^{(j)} < \infty$ and $a = 1$. If $\Phi(x^{(i)}(t)) > \Phi(x^{(j)}(t))$, then robot i swaps its temperature with robot j ; otherwise, they swap with a probability $s(i, j) \in (0, 1)$, which diminishes monotonically with respect to the difference between two objective values. For example, as $\Phi(x^{(i)}) \rightarrow 0$ and $\Phi(x^{(j)}) \rightarrow +\infty$, the swapping rate $s \rightarrow 0$.

Under the specific swapping rule defined in Equation (9), the algorithm exhibits an acceleration effect in concentrating around the global minima of $\Phi(x)$. This effect is theoretically analysed and demonstrated in [40]. Meanwhile, sampling in this manner still leads to the minimization of potential function and hence the identification of the most probable gas source location. This is because the equilibrium distribution of the replica exchange Langevin diffusion is also Equation (8), the same as algorithms in which robots follow independent diffusion processes in Equation (7).

7 Implementation and Evaluation

In this section, we evaluate the system's performance from two aspects: source localization effectiveness and search efficiency. We first describe the settings of both physical and simulation experiments, as well as the metrics used in the evaluation (Section 7.1). We then compare overall performance of *SniffySquad* in both a testbed and simulation (Section 7.2). We also carry out experiments under different conditions to evaluate *SniffySquad*'s robustness to both external and internal factors (Section 7.3).

7.1 Implementation and Setting

7.1.1 Field Experiment Setup. We first describe how we build an environment with gas, then mobile robots deployed in the testbed.

Gas environment. As illustrated in Figure 5, we build a gas leakage testbed in a rectangular $15m \times 10m$ laboratory environment. Generating a gas plume in an enclosed laboratory environment poses strict safety and environmental considerations, making the use of toxic or flammable gases impractical. Our setup employs ethylene glycol (EINECS No. 246-770-3) to achieve a balance between safety and physical representativeness. The behavior of such plumes mirrors that of numerous gas plumes in real-world application scenarios [43, 44]. We simulate the gas leakage scenario by adopting a smoke machine as the gas generator to release ethylene glycol vapor. This choice is consistent with established practice in olfactory robot-based GSL works [27, 45–48]. Among existing works containing physical experiments, commonly used alternatives to toxic or flammable

Table 3. Configuration of Gas Dispersion Parameters

Symbol	Value	Description
μ	10 <i>ppm</i>	Gas conc. at filament center
σ	10 <i>cm</i>	Initial std of filament
γ	10 cm^2/s	Growth ratio of σ
P, T	1 Atm, 298 Kelvins	Pressure and temperature
ν	$1.529 \times 10^{-5} \text{m}^2/\text{s}$	Kinematic viscosity
ρ	$1.196 \text{kg}/\text{m}^3$	Air density
k	$3.75 \times 10^{-3} \text{m}^2/\text{s}^2$	Turb. kinetic energy
ϵ	$1.25 \times 10^{-2} \text{m}^2/\text{s}^3$	Dissipation rate

gases are humid air (using a humidifier as a plume generator) [27], ethanol solution [47, 48], and acetone [49], while many other works conduct only simulation-based validation [50]. We deploy a fan behind the gas generator to generate turbulent wind.

Multi-robot platform. We implement *SniffySquad* on commercial unmanned ground vehicles, as shown in Figure 5(b). Each robot is equipped with a particle matter sensor [51, 52], Plantower PMS5003T [53], which measures the concentration of particles with diameters smaller than three micrometers, expressed in units of micrograms per cubic meter ($\mu\text{g}/\text{m}^3$). When ethylene glycol is atomized, it forms aerosol particles. The particle matter sensor is utilized to detect these particles, allowing us to measure the concentration of ethylene glycol. Additionally, an ultrasonic anemometer has been integrated to provide data on wind direction and speed. To establish ground truth positions, each robot is also equipped with a UWB localization module. Robot movement is guided by a PID controller. The onboard computer of the robot is a Raspberry Pi 4 Computer Model B with 8 GB RAM, and the edge server is equipped with Intel(R) Core(TM) i7-11700 of 2.50 GHz main frequency and 16 G RAM, running Ubuntu 20.04.4. Mobile robots depart from the opposite side of the gas source, i.e., the right side of Figure 5(a).

7.1.2 Simulation Setting. To simulate gas dispersion processes, we follow previous works [35] and build up the simulation environment based on two tools: (i) OpenFOAM [54], a CFD simulator, is used to generate the wind flow vector field; (ii) GADEN [55], a 3-D gas dispersion simulator widely recognized and commonly used for robotic olfaction, is used to generate the gas concentration field. The gas simulation process involves two key steps: First, OpenFOAM models the flow of wind based on the environment geometry, that is, configured by a **computer-aided design (CAD)** model, and the specified flow inlets and outlets; Second, based on the wind flow data, GADEN takes the environment CAD model as input and obtains 3D gas distributions by implementing the filament gas dispersion theory [56]. The specific parameter configuration is listed in Table 3. We fixed parameter settings in our simulator, including gas filaments' center concentration, filaments' initial standard deviation and growth ratio, pressure, temperature, air density, and kinematic viscosity. Our experiments include two representative scenarios in industrial factories: (i) empty spaces Figure 9(a) and (ii) environments with rooms featuring walls and obstacles Figure 9(b). To simulate robot's dynamics, we implement a controller based on [57].

7.1.3 Comparative Methods. We compared *SniffySquad* with two related SOTA methods including: (i) **Surge-Cast** [10], a SOTA reactive bio-inspired GSL algorithm. This algorithm involves two distinct motions patterns: when the robot identifies the plume concentration exceeding a predefined threshold, it engages in a "surge", i.e., moving upwind; otherwise, it executes a "cast" maneuver perpendicular to the wind direction to re-establish contact with the plume; (ii) **Infotaxis** [35], a SOTA probabilistic source parameter estimation-based GSL algorithm. It is an information-driven path

Table 4. Overall performance of field experiment

	SR	PE(%)
SniffySquad	5/6	90.0
Infotaxis	4/6	62.0
Surge-Cast	4/6	61.0

planning algorithm, leveraging a probability map of the source position estimation and driving the robot's movement toward the place with the highest information gain. To ensure fair comparisons, all evaluated methods, including Surge-Cast, Infotaxis, and ours, were implemented and tested under identical hardware and environmental conditions.

We excluded source seeking methods based on **deep reinforcement learning (DRL)** because their reliance on offline training in a specific environment geometry contradicts the requirement for a rapid-response system in emergency scenarios. For example, in CATNIPP [29], the agent learns a subtle global representation of its entire belief over the domain, enabling it to sequence non-myopic decisions that can achieve longer-term objectives. However, the policy learned via DRL is intrinsically environment-specific. It is tailored to the geometry used during training and would fail or require retraining when deployed in a new environment. Such methods would be applicable to scenarios like coral head localization with an **autonomous surface vehicle (ASV)** [58], or elevation/bathymetric mapping using an ASV with a single-beam sonar pointing downward to collect depth measurement [57]. However, in our context of indoor, factory-scale gas dispersion, the concentration field is dictated by the environment's specific geometry.

7.1.4 Evaluation Metrics. We employ the following two metrics to evaluate the algorithms' effectiveness and efficiency, respectively: (i) **Success Rate (SR)**. It is determined by calculating the ratio of successful trials across all repeated experiments. A trial is deemed successful if mobile olfactory robots reach the vicinity of the actual ground truth position of the gas source within the predefined time limit and distance threshold ε_s . The threshold is set to $0.5m$ in the experiments. In practical application scenarios, the gas leakage source can be detected using onboard cameras as soon as it comes within a robot's line-of-sight. (ii) **Path Efficiency (PE)**. It is defined as $\frac{d_{min}}{d}$, where d_{min} is the shortest path distance from the robot's starting point to the gas source, and d is the length of the robot's travel trajectory driven by GSL approaches. This metric is only calculated for successful trials. Note that PE is proportional to the time required for localizing the gas source, which aligns with the objective in Equation (1).

7.2 Overall Performance

Table 4 illustrates overall performance evaluated in the testbed. A total of six experiments were conducted to compare the performance between Surge-Cast to Infotaxis and *SniffySquad*. In terms of both SR and PE, the performance of *SniffySquad* surpasses two SOTA baselines, Surge-Cast and Infotaxis. *SniffySquad* improves the PE (i.e., decreases the trajectory lengths) by 32% and 32% compared to Surge-Cast and Infotaxis, respectively.

Figure 6 evaluates overall performance of *SniffySquad* in physical feature-based simulations, using various numbers of robots. These experiments were carried out 50 times for each configuration. Across all scenarios involving robot counts ranging from one to three, it is evident that *SniffySquad* consistently attains the highest SR, achieves the shortest search time for both an individual robot and robot teams, and generates the most efficient path. Specifically, *SniffySquad* achieves a SR of 76% for a team of three robots, improving by 30% and 20% in comparison with

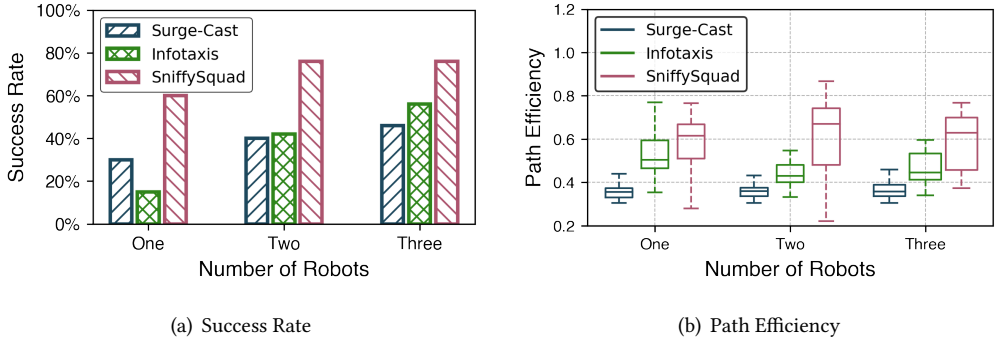


Fig. 6. Overall performance of physical feature-based simulation experiments.

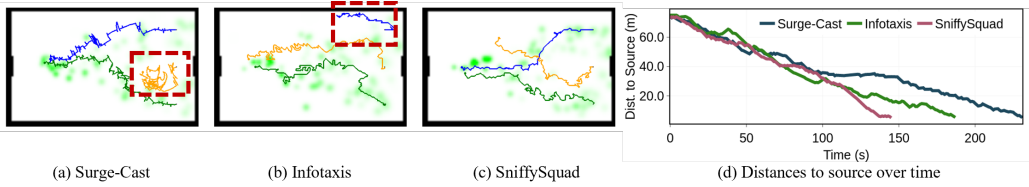


Fig. 7. In an empty space (w/o walls and obstacles). (a)–(c) Illustration of trajectories generated by Surge-Cast, Infotaxis, and *SniffySquad*. The red rectangles indicate the trajectories that were misled by the gas patches. (d) The distance of the robot to the gas release source over time.

Surge-Cast and Infotaxis, respectively. The search time and PE are 148s and 0.67, reducing the total search time by 22.92% and 18.46%, shortening the trajectory length by 41.79% (from $2.56d_{min}$ to $1.49d_{min}$) and 32.84% (from $2.22d_{min}$ to $1.49d_{min}$), respectively. This results from the resilience brought by the individual-level movement planning and the team-level roles adaptation algorithm, which enable robots to escape from false positive sources and parallelize the exploration of the environment and exploitation of collected information in a flexible manner. For all algorithms, SR show an upward trend as the number of robots employed for gas source search and localization increases. Apparently, this is because the area is explored to a greater extent when more robots are working.

To further compare the performance of *SniffySquad* and baseline methods, we visualize the trajectories of all methods in Figure 7(a)–(c). It is demonstrated that *SniffySquad* produces smoother trajectories than baselines. Surge-Cast, a reactive method relying solely on current observations to make decisions, yields inefficient zig-zag paths and is the most susceptible to spatial and temporal gas concentration fluctuations (as shown in Figure 2). Infotaxis, though incorporating a Bayesian gas source position estimator to filter the sensory measurements, suffers from trap in patchy areas. Figure 7(d) illustrates the nearest distance to the source within all robots over time. It is verified that *SniffySquad* navigates towards the source the most efficiently and arrives at the source the most quickly. Surge-Cast takes 231 seconds and Infotaxis takes 187 seconds in total while *SniffySquad* takes only 145 seconds, indicating a decrease in search time by 37.23% and 22.46% in comparison to the two baselines, respectively. The acceleration effect produced by *SniffySquad* relative to previous methods in localizing the source attributes to the movement planning method that regulates the robots' moving direction to enable escaping from gas patches, as well as the collaborative assignment and adaptation of team roles.

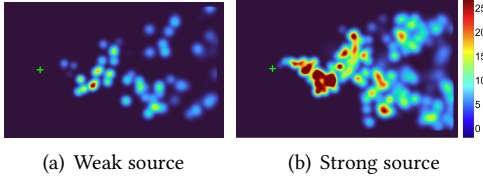


Fig. 8. Illustration of gas concentration fields under different emission intensities.

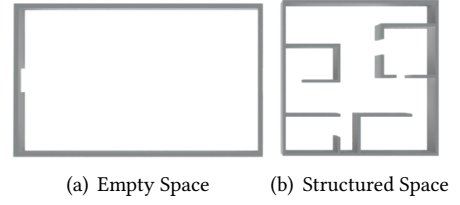


Fig. 9. Illustration of environment geometries.

7.3 System Robustness

We further experimentally evaluate *SniffySquad* with respect to various environmental conditions and key parameter settings in our system.

7.3.1 Impact of Gas Emission Intensity. In real-world scenarios, *SniffySquad* should be capable of handling situations where different gas emission intensities are presented. Therefore, we investigate the performance of our system when handling different release rates of the gas source. Specifically, we select two representative gas release rates of 10 filaments per second and 30 filaments per second, which are denoted as a weak source and a strong source, respectively. As illustrated in Figure 8(a) and 8(b), a snapshot of the field with the weak source shows a more discontinuous gas concentration field compared to that of the strong source, which indicates a higher difficulty for GSL.

We compare the performance of the proposed method and baselines under these two scenarios, as shown in Figure 10. Firstly, across scenarios of a strong and a weak source, our method consistently achieves the highest SR and the most efficient paths among all methods. This can be attributed to the patchy plume-resilient movement planning mechanism, as well as role adaptation based collaboration among robots in the team. Secondly, for all the methods, the SR and PE degrade as the gas emission intensity from strong turns to weak. This is because the olfactory stimulus in a field induced by the weaker source is sparser, confusing the robots more severely. Thirdly, it is noteworthy that our approach's advantage is especially significant under the weak gas source, validating its remarkable robustness to gas emission intensity. From the strong to weak source scenario, Surge-Cast and Infotaxis' SR decrease considerably by 12% and 20%, respectively, while their PE degrade significantly by 41% and 33%, respectively. In contrast, *SniffySquad* still maintains a SR of above 89% and its PE slightly reduces by only 13%. This is because of the integrated design of sensing and planning processes in our method, which allows the robot to escape from spurious gas patches and leads to more efficient localization.

7.3.2 Impact of Environmental Geometry. We evaluate the robustness of our approach in different environment geometries. As shown in Figure 9(a) and 9(b), we evaluate all the methods in an empty space (without obstacles) and a structured space (with walls and rooms) to verify the benefits of our method. From Figure 11, we can see that *SniffySquad* keeps outperforming the baseline method on both the SR and PE. Besides, although the structured space significantly increases the difficulty of GSL, our method maintains a SR of more than 70% and an average PE of around 0.6%. Note that Infotaxis shows a better performance than Surge-Cast in the empty space, but its performance degrades greatly when deployed in a more complex structured space. This indicates that *SniffySquad* avoids the oversimplified assumption of either the gas model or the environment, thus featuring the capability to handle complex environments with less performance degradation. While demonstrating improved performance in handling plume patchiness compared to baseline

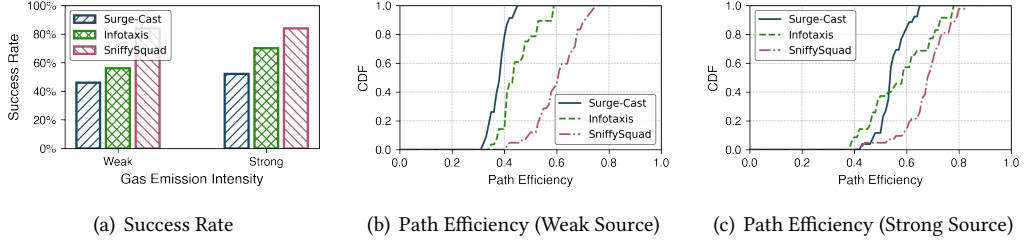


Fig. 10. Impact of gas emission intensity. (a) SR. (b)–(c) CDF of PE under a weak and a strong source, respectively.

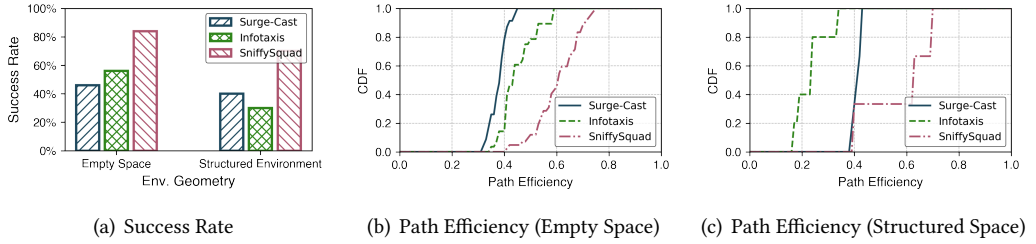


Fig. 11. Impact of environment geometry. (a) SR. (b)–(c) CDF of PE under an empty and a structured space, respectively.

methods, we also acknowledge that remaining limitations and areas for improvement exist for structured environments.

7.3.3 Impact of Initial Distance to the Source. We examine the impact of robots' initial distances to the gas source on the task. The robots initiate their journey from five distinct positions, ranging from close to gradually increasing distances relative to the gas source. As shown in Figure 12, when the initial distance is less than 70m, the SR remains around 100% and the PE decreases gradually as the initial distance grows. However, when it comes to a distance larger than 70m, we notice that the SR quickly decreases. This is because the gas plume becomes more and more patchy as the distance increases (as illustrated in Figure 2), inducing higher probability that the robots are entrapped or misled to a false source localization. We also note that even when the robots start searching at 90m away from the source, *SniffySquad* can still achieve an average search time of 148.1s for the successful trials.

7.3.4 Impact of Temperature Parameters. As clarified in Section 6, the temperature τ is a critical parameter in the system as it governs the role of a robot. To clarify the impact of temperature parameters on the system performance, all robots are assigned an identical temperature across five orders of magnitude in the experiments. These experiments are conducted in an empty space (without obstacles) as shown in Figure 9(a). As illustrated in Figure 13, the results of SR and PE exhibit different trends as the temperature rises. For the SR, as the temperature increases, it shows an increasing trend at first and reaches the maximum at the temperature of 10^{-1} , and then declines gradually; for the PE, its value gradually decreases as the temperature grows, especially when it rises over 10^{-1} . This pattern echoes our previous analysis that either too exploitative movement (low temperature) or explorative movement (high temperature) fails to balance the trade-off of search efficiency and effectiveness. *SniffySquad* combines the advantages of both

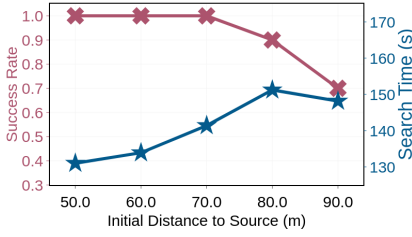


Fig. 12. Impact of initial distance to source.

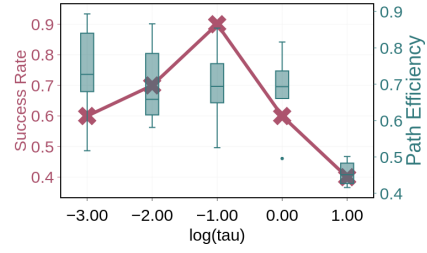


Fig. 13. Impact of temperature parameter.

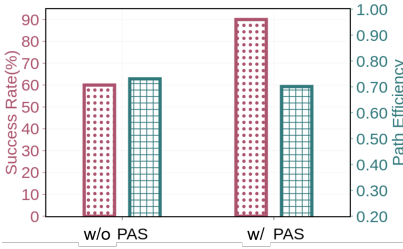


Fig. 14. Effectiveness of patchiness-aware active sensing strategy.

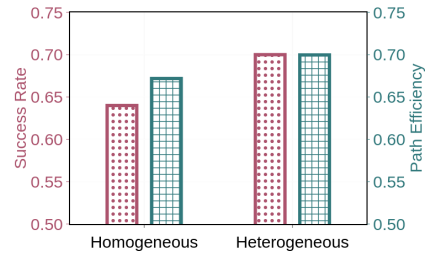


Fig. 15. Effectiveness of the collaborative roles adaptation strategy.

behaviors via the collaborative role-switching method, making *SniffySquad* more superior to other approaches.

7.4 System Micro-benchmark

We further conduct an ablation study to understand the effectiveness of components introduced in Sections 5 and 6.

7.4.1 Effectiveness of the Patchiness-aware Active Sensing. In this experiment, we compare the GSL performance without (w/o) and with (w/) the proposed Patchiness-aware Active Sensing component to show the performance gains it brings into the overall system. The results are shown in Figure 14. As seen, with our movement planning algorithm, the system achieves an enhanced performance, where the SR increases without sacrificing the PE. Specifically, the SR of *SniffySquad* with the novel motion strategy is 0.9, increasing that of *SniffySquad* without the motion strategy by $\sim 30\%$. Meanwhile, the trajectory efficiency is decreased by merely 4.3%.

7.4.2 Effectiveness of the Collaborative Roles Adaptation strategy. To demonstrate the effectiveness of the Collaborative Role Adaptation module in the *SniffySquad*, we evaluate the localization effectiveness and search efficiency with and without assigning different roles and adaptively adjusting them through Equation 9. The results are shown in Figure 15. We divide the teams of robots into two categories, homogeneous and heterogeneous, where robots are acting the same role and different roles, respectively. Specifically, heterogeneous teams consist of all combinations in which at least one robot acts a different role from others, and the metrics are averaged for experiments using a range of temperature values. The SR and trajectory length (PE) improve from 0.64 and $1.49d_{min}$ to 0.7 and $1.43d_{min}$, respectively. The approach without the Role Assignment and Adjustment module is surpassed by the approach with the component, utilizing the complementary capabilities of multiple robots and further exploits them through adaptive role swapping. Specifically, this is attributed to the distinct temperatures representing diverse roles within the team, which forms

the basis of the role exchange mechanism in the proposed method. This underscores that flexible allocation of roles within a team enhances agile exploration or exploitation of robots, effectively expediting global-scale search while mitigating local optima concerns.

8 Discussion

8.1 System Resource Analysis

8.1.1 Computational cost. We analyze the computational complexity of the two main processes in our system: global map maintenance and the decentralized coordination strategy.

Computational cost of global probability map maintenance. The computational complexity of the probabilistic gas source estimation (Section 5.3) scales as $\mathcal{O}(NM)$, where N represents the number of grid cells in the map and M is the number of robots. Mapping, as a fundamental task, has been widely considered in robotic applications, such as coverage path planning [59] and **Simultaneous Localization and Mapping (SLAM)** [60]. In tasks like SLAM, robots typically have to maintain a map that consumes much more memory to store complex and high-dimensional visual features. There are several favored ways in multi-robot systems to reduce the computational costs associated with mapping. For example: (a) Efficient data structures: adopting a storage-efficient data structure rather than a fixed grid, such as a **random finite set (RFS)** maintaining a hierarchical decomposition of the environment [61]. (b) Graph sparsification: this aims to approximate the complete graph with as few edges as possible, mainly by removing redundant edges that are not providing new information during the estimation process [60].

Computational cost of roles adaptation coordination. The team roles adaptation strategy (presented in Algorithm 1) evaluates the swapping rate for a potential role exchange by comparing each robot with every other team member during the coordination step. This results in a complexity of $\mathcal{O}(M^2)$ for the coordination strategy, where M is the number of robots.

8.1.2 Communication Load and Frequency. First, we analyze the size of packets communicated between the robots and the server. Communication occurs in two directions under two circumstances:

- (1) **Server-to-Robot Packets (Commands).** Packets from central server to robots are used for requesting sensing data or sending motion commands. The server's motion commands' format and content depend on the robot's kinematics. In our evaluation, we used Mecanum wheel vehicles, whose motion commands contain a 2-D velocity vector, rotating velocity, and the time duration.
- (2) **Robots-to-Server Packets (Sensing/Position).** Upon sensing requests from the server, robots send a packet containing sensing data (the gas intensity, wind speed, wind direction), and positional data (the robot's coordinates and heading).

Under both circumstances (1) and (2), the packet contains less than eight floating-point numbers. We used the default 64-bit float type in Python for numerical data, hence the data payload of a packet is merely dozens of bytes.

Next, we analyze the communication frequency. In both circumstances (1) and (2) introduced above, communication happens for two purposes: motion control and sensing.

- **Motion Control Frequency:** The communication required for continuous low-level motion control is set as a frequency of 10 Hz.
- **Sensing and Planning Frequency:** Communication for sensing is triggered at the start of each high-level timestep. It is event-driven, not constant frequency, and occurs at intervals of several seconds. Specifically, in each timestep, the central server begins by requesting sensing data from robots. Based on this data, the server plans the next motion and assigns each robot

a goal position. Robots then navigate to their goals using PID control. When they arrive at the goal, the current timestep ends and the next timestep starts.

8.2 Safety Considerations

(i) *Toxicology*. Laboratory studies have shown that ethylene glycol penetrates animal skin slowly and that absorption through human skin takes place even more slowly [62, 63]. A splash of glycol liquids in the eyes may cause temporary discomfort or irritation, but permanent damage is not likely. (ii) *Flammability*. Ethylene glycol has a relatively high flash point, viz, 115 °C (240 °F) [62], meaning it is not considered highly flammable at room temperature. Moreover, since our solution contains a high proportion of water, which itself has no flash point and acts as a fire suppressant, the overall solution we used in our experiments is non-flammable under normal laboratory conditions. (iii) *Ventilation and Robot Decontamination*. All experiments were conducted in a well-ventilated indoor environment, with either HVAC (Heating, Ventilation, and Air Conditioning) systems or fans providing continuous airflow to ensure that EG solution vapor remained low. The robot platforms were wiped down with water-based cleaning agent on a regular basis after several trials to minimize long-term exposure risks.

9 Conclusion

This article proposes *SniffySquad*, a multi-robot olfaction-based system for GSL. *SniffySquad* features two key designs from nonconvex learning perspective to address the patchy characteristic of gas plumes and enable efficient and effective source localization by a collaborative strategy, as elaborated below. Specifically, we devise a patchy plume-resilient movement strategy inspired by Langevin diffusion approach. We then develop a collaborative strategy for multiple robots by assigning and adapting their planning preferences dynamically to balance the search efficiency and localization effectiveness of gas sources. We implement *SniffySquad* with multiple unmanned ground vehicles and conduct experiments in a real-world testbed. Results show that *SniffySquad* achieves a remarkable 20%+ SR and 30%+ PE improvement, respectively, outperforming SOTA GSL solutions.

References

- [1] Xinlei Chen, Susu Xu, Xinyu Liu, Xiangxiang Xu, Hae Young Noh, Lin Zhang, and Pei Zhang. 2020. Adaptive hybrid model-enabled sensing system (HMSS) for mobile fine-grained air pollution estimation. *IEEE Transactions on Mobile Computing* 21, 6 (2020), 1927–1944.
- [2] Zuxin Li, Fanhang Man, Xuecheng Chen, Susu Xu, Fan Dang, Xiao-Ping Zhang, and Xinlei Chen. 2024. QUEST: Quality-informed multi-agent dispatching system for optimal mobile crowdsensing. In *Proceedings of the IEEE INFOCOM 2024-IEEE Conference on Computer Communications*. IEEE, 1811–1820.
- [3] Sujata Pal, Anindo Ghosh, and Vivek Sethi. 2018. Vehicle air pollution monitoring using iots. In *Proceedings of the 16th ACM Conference on Embedded Networked Sensor Systems*. 400–401.
- [4] Balz Maag, Zimu Zhou, and Lothar Thiele. 2018. W-air: Enabling personal air pollution monitoring on wearables. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 1 (2018), 1–25.
- [5] Xue Sun, Jie Xiong, Chao Feng, Xiaohui Li, Jiayi Zhang, Binghao Li, Dingyi Fang, and Xiaojiang Chen. 2024. Gastag: A gas sensing paradigm using graphene-based tags. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*. 342–356.
- [6] Geralyn McCarron. 2018. Air pollution and human health hazards: A compilation of air toxins acknowledged by the gas industry in Queensland’s Darling Downs. *International Journal of Environmental Studies* 75, 1 (2018), 171–185.
- [7] Gianfranco Manes, Giovanni Collodi, Leonardo Gelpi, Rosanna Fusco, Giuseppe Ricci, Antonio Manes, and Marco Passafiume. 2016. Realtime gas emission monitoring at hazardous sites using a distributed point-source sensing infrastructure. *Sensors* 16, 1 (2016), 121.
- [8] Joseph R. Bourne, Matthew N. Goodell, Xiang He, Jake A. Steiner, and Kam K. Leang. 2020. Decentralized multi-agent information-theoretic control for target estimation and localization: Finding gas leaks. *The International Journal of Robotics Research* 39, 13 (2020), 1525–1548.

- [9] Michael Hutchinson, Hyondong Oh, and Wen-Hua Chen. 2018. Entrotaxis as a strategy for autonomous search and source reconstruction in turbulent conditions. *Information Fusion* 42 (2018), 179–189.
- [10] Gautam Reddy, Venkatesh N. Murthy, and Massimo Vergassola. 2022. Olfactory sensing and navigation in turbulent environments. *Annual Review of Condensed Matter Physics* 13 (2022), 191–213.
- [11] Xin-xing Chen and Jian Huang. 2019. Odor source localization algorithms on mobile robots: A review and future outlook. *Robotics and Autonomous Systems* 112 (2019), 123–136.
- [12] Callum Rhodes, Cunjia Liu, Paul Westoby, and Wen-Hua Chen. 2023. Autonomous search of an airborne release in urban environments using informed tree planning. *Autonomous Robots* 47, 1 (2023), 1–18.
- [13] Callum Rhodes, Cunjia Liu, and Wen-Hua Chen. 2020. Informative path planning for gas distribution mapping in cluttered environments. In *Proceedings of the 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 6726–6732.
- [14] Joseph R. Bourne, Eric R. Pardyjak, and Kam K. Leang. 2019. Coordinated Bayesian-based bioinspired plume source term estimation and source seeking for mobile robots. *IEEE Transactions on Robotics* 35, 4 (2019), 967–986.
- [15] Bardienus P. Duisterhof, Shushuai Li, Javier Burgués, Vijay Janapa Reddi, and Guido CHE de Croon. 2021. Sniffy bug: A fully autonomous swarm of gas-seeking nano quadcopters in cluttered environments. In *Proceedings of the 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 9099–9106.
- [16] Wei Deng, Guang Lin, and Faming Liang. 2020. A contour stochastic gradient langevin dynamics algorithm for simulations of multi-modal distributions. *Advances in Neural Information Processing Systems* 33 (2020), 15725–15736.
- [17] Aurore Loisy and Christophe Eloy. 2022. Searching for a source without gradients: how good is infotaxis and how to beat it. *Proceedings of the Royal Society A* 478, 2262 (2022), 20220118.
- [18] Ekkehard Holzbecher and Ekkehard Holzbecher. 2012. 2D and 3D transport solutions (Gaussian Puffs and Plumes). *Environmental Modeling: Using MATLAB* (2012), 303–316.
- [19] Xuecheng Chen, Haoyang Wang, Yuhan Cheng, Haohao Fu, Yuxuan Liu, Fan Dang, Yunhao Liu, Jinqiang Cui, and Xinlei Chen. 2024. Ddl: Empowering delivery drones with large-scale urban sensing capability. *IEEE Journal of Selected Topics in Signal Processing* 18, 3 (2024), 502–515.
- [20] Xuecheng Chen, Zijian Xiao, Yuhan Cheng, Chen-Chun Hsia, Haoyang Wang, Jingao Xu, Susu Xu, Fan Dang, Xiao-Ping Zhang, Yunhao Liu, et al. 2024. Soscheduler: Toward proactive and adaptive wildfire suppression via multi-uav collaborative scheduling. *IEEE Internet of Things Journal* 11, 14 (2024), 24858–24871.
- [21] Xinlei Chen, Xiangxiang Xu, Xinyu Liu, Shijia Pan, Jiayou He, Hae Young Noh, Lin Zhang, and Pei Zhang. 2018. Pga: Physics guided and adaptive approach for mobile fine-grained air pollution estimation. In *Proceedings of the 2018 ACM International Joint Conference and 2018 International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers*. 1321–1330.
- [22] Nan Zhou, Yuxuan Liu, Haoyang Wang, Fanhang Man, Jingao Xu, Fan Dang, Chaopeng Hong, Yunhao Liu, Xiao-Ping Zhang, Yali Song, et al. 2025. CatUA: Catalyzing urban air quality intelligence through mobile crowd-sensing. *IEEE Transactions on Mobile Computing* 24, 9 (2025), 9184–9201.
- [23] Massimo Vergassola, Emmanuel Villermaux, and Boris I Shraiman. 2007. ‘Infotaxis’ as a strategy for searching without gradients. *Nature* 445, 7126 (2007), 406–409.
- [24] Chaoqun Wang, Teng Li, Max Q.-H. Meng, and Clarence De Silva. 2018. Efficient mobile robot exploration with Gaussian Markov random fields in 3D environments. In *Proceedings of the 2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 5015–5021.
- [25] Achim J. Lilienthal, Matteo Reggente, Marco Trincavelli, Jose Luis Blanco, and Javier Gonzalez. 2009. A statistical approach to gas distribution modelling with mobile robots-the kernel dm+ v algorithm. In *Proceedings of the 2009 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 570–576.
- [26] Andres Gongora, Javier Monroy, and Javier Gonzalez-Jimenez. 2020. Joint estimation of gas and wind maps for fast-response applications. *Applied Mathematical Modelling* 87 (2020), 655–674.
- [27] Joseph R. Bourne, Eric R. Pardyjak, and Kam K. Leang. 2019. Coordinated Bayesian-based bioinspired plume source term estimation and source seeking for mobile robots. *IEEE Transactions on Robotics* 35, 4 (2019), 967–986.
- [28] Lino Marques, Urbano Nunes, and Anibal T. de Almeida. 2002. Olfaction-based mobile robot navigation. *Thin Solid Films* 418, 1 (2002), 51–58. DOI: [http://doi.org/10.1016/S0040-6090\(02\)00593-X](http://doi.org/10.1016/S0040-6090(02)00593-X) Proceedings from the International School on Gas Sensors in conjunction with the 3rd European School of the NOSE Network.
- [29] Yuhong Cao, Yizhuo Wang, Apoorva Vashisth, Haolin Fan, and Guillaume Adrien Sartorette. 2023. CatNIPP: Context-aware attention-based network for informative path planning. In *Proceedings of the Conference on Robot Learning*. PMLR, 1928–1937.
- [30] Haitong Ma, Tianpeng Zhang, Yixuan Wu, Flávio du Pin Calmon, and Na Li. 2023. Gaussian max-value entropy search for multi-agent bayesian optimization. *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2023), 10028–10035.

- [31] Tianpeng Zhang, Victor Qin, Yujie Tang, and Na Li. 2022. Distributed information-based source seeking. *IEEE Transactions on Robotics* 39, 6 (2022), 4749–4767. Retrieved from <https://api.semanticscholar.org/CorpusID:252383539>
- [32] Qiuyi Gu, Zhaocheng Ye, Jincheng Yu, Jiahao Tang, Tinghao Yi, Yuhan Dong, Jian Wang, Jinqiang Cui, Xinlei Chen, and Yu Wang. 2025. MR-COGraphs: Communication-efficient multi-robot open-vocabulary mapping system via 3D scene graphs. *IEEE Robotics and Automation Letters* 10, 6 (2025), 5713–5720.
- [33] Xinlei Chen, Susu Xu, Jun Han, Haohao Fu, Xidong Pi, Carlee Joe-Wong, Yong Li, Lin Zhang, Hae Young Noh, and Pei Zhang. 2020. Pas: Prediction-based actuation system for city-scale ridesharing vehicular mobile crowdsensing. *IEEE Internet of Things Journal* 7, 5 (2020), 3719–3734.
- [34] Thomas Lochmatter and Alcherio Martinoli. 2009. Tracking odor plumes in a laminar wind field with bio-inspired algorithms. In *Proceedings of the Experimental Robotics: The Eleventh International Symposium*. Springer, 473–482.
- [35] Pepe Ojeda, Javier Monroy, and Javier Gonzalez-Jimenez. 2021. Information-driven gas source localization exploiting gas and wind local measurements for autonomous mobile robots. *IEEE Robotics and Automation Letters* 6, 2 (2021), 1320–1326.
- [36] Qilin Feng, Canxin Zhang, Jingyu Lu, Hao Cai, Zhilong Chen, Yibin Yang, Fei Li, and Xianting Li. 2019. Source localization in dynamic indoor environments with natural ventilation: an experimental study of a particle swarm optimization-based multi-robot olfaction method. *Building and Environment* 161 (2019), 106228.
- [37] Diederik P. Kingma and Jimmy Ba. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*. Retrieved from <https://arxiv.org/abs/1412.6980> (2014).
- [38] Xiang Cheng. 2020. *The Interplay between Sampling and Optimization*. University of California, Berkeley.
- [39] Yi-An Ma, Yuansi Chen, Chi Jin, Nicolas Flammarion, and Michael I. Jordan. 2019. Sampling can be faster than optimization. *Proceedings of the National Academy of Sciences* 116, 42 (2019), 20881–20885.
- [40] Yi Chen, Jinglin Chen, Jing Dong, Jian Peng, and Zhaoran Wang. 2020. Accelerating nonconvex learning via replica exchange Langevin diffusion. *arXiv preprint arXiv:2007.01990*. Retrieved from <https://arxiv.org/abs/2007.01990> (2020).
- [41] John C. Strikwerda. 2004. *Finite difference schemes and partial differential equations*. SIAM.
- [42] David J. Earl and Michael W. Deem. 2005. Parallel tempering: Theory, applications, and new perspectives. *Physical Chemistry Chemical Physics* 7, 23 (2005), 3910–3916.
- [43] Nirag Kadakia, Mahmut Demir, Brenden T. Michaelis, Brian D. DeAngelis, Matthew A. Reidenbach, Damon A. Clark, and Thierry Emonet. 2022. Odour motion sensing enhances navigation of complex plumes. *Nature* 611, 7937 (2022), 754–761.
- [44] Adam Francis, Shuai Li, Christian Griffiths, and Johann Sienz. 2022. Gas source localization and mapping with mobile robots: A review. *Journal of Field Robotics* 39, 8 (2022), 1341–1373.
- [45] Michael Hutchinson, Pawel Ladosz, Cunjia Liu, and Wen-Hua Chen. 2019. Experimental assessment of plume mapping using point measurements from unmanned vehicles. In *Proceedings of the 2019 International Conference on Robotics and Automation (ICRA)*. IEEE, 7720–7726.
- [46] Michael Hutchinson, Cunjia Liu, and Wen-Hua Chen. 2018. Information-based search for an atmospheric release using a mobile robot: Algorithm and experiments. *IEEE Transactions on Control Systems Technology* 27, 6 (2018), 2388–2402.
- [47] Wanting Jin and Alcherio Martinoli. 2024. Sense in motion with belief clustering: efficient gas source localization with mobile robots. In *Proceedings of the 2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 14909–14916.
- [48] Pepe Ojeda, Javier Monroy, and Javier Gonzalez-Jimenez. 2024. Robotic gas source localization with probabilistic mapping and online dispersion simulation. *IEEE Transactions on Robotics* 40 (2024), 3551–3564.
- [49] Michael Hutchinson, Cunjia Liu, and Wen-Hua Chen. 2019. Source term estimation of a hazardous airborne release using an unmanned aerial vehicle. *Journal of Field Robotics* 36, 4 (2019), 797–817.
- [50] Pepe Ojeda, Javier Monroy, and Javier Gonzalez-Jimenez. 2025. PSGSL: A probabilistic framework integrating semantic scene understanding and gas sensing for gas source localization. *arXiv preprint arXiv:2501.12812*. Retrieved from <https://arxiv.org/abs/2501.12812> (2025).
- [51] Yuxuan Liu, Haoyang Wang, Fanhang Man, Jingao Xu, Fan Dang, Yunhao Liu, Xiao-Ping Zhang, and Xinlei Chen. 2024. Mobaiir: Unleashing sensor mobility for city-scale and fine-grained air-quality monitoring with Airbert. In *Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services*. 223–236.
- [52] Ji Luo, Zijian Xiao, Zuxin Li, Xuecheng Chen, Chaopeng Hong, Xiao-Ping Zhang, and Xinlei Chen. 2025. SmartSpr: A physics-informed mobile sprinkler scheduling system for reducing urban particulate matter pollution. *IEEE Transactions on Mobile Computing* 24, 9 (2025), 7994–8010.
- [53] Misti Levy Zamora, Fulizi Xiong, Drew Gentner, Branko Kerkez, Joseph Kohrman-Glaser, and Kirsten Koehler. 2018. Field and laboratory evaluations of the low-cost plantower particulate matter sensor. *Environmental science & technology* 53, 2 (2018), 838–849.
- [54] Hrvoje Jasak. 2009. OpenFOAM: Open source CFD in research and industry. *International Journal of Naval Architecture and Ocean Engineering* 1, 2 (2009), 89–94.

- [55] Javier Monroy, Victor Hernandez-Bennetts, Han Fan, Achim Lilienthal, and Javier Gonzalez-Jimenez. 2017. GADEN: A 3D gas dispersion simulator for mobile robot olfaction in realistic environments. *Sensors* 17, 7 (2017), 1479.
- [56] Jay A. Farrell, John Murlis, Xuezhong Long, WEI Li, and Ring T. Cardé. 2002. Filament-based atmospheric dispersion model to achieve short time-scale structure of odor plumes. *Environmental fluid mechanics* 2, 1 (2002), 143–169.
- [57] Weizhe Chen, Roni Khardon, and Lantao Liu. 2024. Adaptive robotic information gathering via non-stationary Gaussian processes. *The International Journal of Robotics Research* 43, 4 (2024), 405–436.
- [58] Genevieve Flaspohler, Victoria Preston, Anna P. M. Michel, Yogesh Girdhar, and Nicholas Roy. 2019. Information-guided robotic maximum seek-and-sample in partially observable continuous environments. *IEEE Robotics and Automation Letters* 4, 4 (2019), 3782–3789.
- [59] Leighton Collins, Payam Ghassemi, Ehsan T Esfahani, David Doermann, Karthik Dantu, and Souma Chowdhury. 2021. Scalable coverage path planning of multi-robot teams for monitoring non-convex areas. In *Proceedings of the 2021 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 7393–7399.
- [60] Pierre-Yves Lajoie and Giovanni Beltrame. 2023. Swarm-slam: Sparse decentralized collaborative simultaneous localization and mapping framework for multi-robot systems. *IEEE Robotics and Automation Letters* 9, 1 (2023), 475–482.
- [61] Philip M. Dames, Mac Schwager, Daniela Rus, and Vijay Kumar. 2015. Active magnetic anomaly detection using multiple micro aerial vehicles. *IEEE Robotics and Automation Letters* 1, 1 (2015), 153–160.
- [62] Robert Fulton Dye. 2001. Ethylene glycols technology. *Korean Journal of Chemical Engineering* 18, 5 (2001), 571–579.
- [63] J. S. LaKind, EA McKenna, R. P. Hubner, and R. G. Tardiff. 1999. A review of the comparative mammalian toxicity of ethylene glycol and propylene glycol. *Critical Reviews in Toxicology* 29, 4 (1999), 331–365.
- [64] Raimondas Galvelis and Yuji Sugita. 2015. Replica state exchange metadynamics for improving the convergence of free energy estimates. *Journal of Computational Chemistry* 36, 19 (2015), 1446–1455.
- [65] Guanxun Li, Guang Lin, Zecheng Zhang, and Quan Zhou. 2023. Fast replica exchange stochastic gradient langevin dynamics. *arXiv preprint arXiv:2301.01898*. Retrieved from <https://arxiv.org/abs/2301.01898> (2023).
- [66] Herbert Robbins and Sutton Monro. 1951. A stochastic approximation method. *The Annals of Mathematical Statistics* 22, 3 (1951), 400–407.
- [67] Lauren A. Hannah. 2015. Stochastic optimization. *International Encyclopedia of the Social & Behavioral Sciences* 2 (2015), 473–481.
- [68] Zi Wang and Stefanie Jegelka. 2017. Max-value entropy search for efficient Bayesian optimization. In *Proceedings of the International Conference on Machine Learning*. PMLR, 3627–3635.

Appendices

A Theoretical Analysis of Section 6 Collaborative Roles Adaptation

A.1 Swapping Rate Formula

The form of the swapping rate in Equation (9) arises from the Metropolis-Hastings algorithm [64]. Consider a move where the configurations of two replicas (i.e., two robots' positions in our problem), $x^{(i)}$ and $x^{(j)}$, are swapped:

$$X = (\dots, x^{(i)}, \dots, x^{(j)}, \dots), \quad X' = (\dots, x^{(j)}, \dots, x^{(i)}, \dots).$$

In the Metropolis-Hastings algorithm, a move (i.e., exchange in our method) is proposed according to some proposal distribution and then accepted with probability $T(X \rightarrow X') = \min\left(1, \frac{\pi(X')q(X' \rightarrow X)}{\pi(X)q(X \rightarrow X')}\right)$, where the stationary distribution is $\pi(X) = \prod_k e^{-\frac{1}{\tau_k} \Phi(x^{(k)})}$. Assuming a symmetric proposal distribution, $q(X \rightarrow X') = q(X' \rightarrow X)$, the Metropolis-Hastings acceptance probability simplifies to

$$T(X \rightarrow X') = \min\left(1, \frac{P(x^{(j)}, x^{(i)})}{P(x^{(i)}, x^{(j)})}\right).$$

Substituting the joint probabilities for the swapped and unswapped configurations, we obtain

$$T(X \rightarrow X') = \min\left(1, e^{\left(\frac{1}{\tau_i} - \frac{1}{\tau_j}\right)(\Phi(x^{(i)}) - \Phi(x^{(j)}))}\right),$$

which corresponds to the swapping rate $s(i, j)$ in Equation 9 of the manuscript.

A.2 Convergence Analysis

The convergence analysis of the collaborative roles adaptation module can be approached by examining the evolution of the χ^2 -divergence [40] $\chi^2(\mu_t \parallel \pi)$, which quantifies the discrepancy between μ_t and π , where μ_t is the distribution of the replica exchange Langevin diffusion at time t , and π is its invariant distribution. Equation (3.8) in [40] provides that:

$$\frac{d}{dt}\chi^2(\mu_t \parallel \pi) = -2 \mathcal{E}^a\left(\frac{d\mu_t}{d\pi}\right),$$

where the Dirichlet form [65] of a fixed function f is given by:

$$\mathcal{E}^a(f) = \mathcal{E}^0(f) + \int \frac{a}{2} s(x^{(1)}, x^{(2)}) \left(f(x^{(2)}, x^{(1)}) - f(x^{(1)}, x^{(2)}) \right)^2 d\pi(x^{(1)}, x^{(2)}).$$

Here, the first term $\mathcal{E}^0(f)$ corresponds to $a = 0$, i.e., the Langevin diffusion without swapping. The second term is nonnegative, which increases the Dirichlet form and thereby accelerates the evolution of $\chi^2(\mu_t \parallel \pi)$ [40].

B Choice of Temperature Parameter

On one hand, a larger temperature parameter τ promotes faster global exploration. On the other hand, if τ is excessively large, exploration becomes fast but ineffective. As discussed in Appendix A of [40], when τ is too large, the gradient term $\nabla\Phi(\cdot)$ becomes negligible compared to the white noise term $\sqrt{2\tau}dB$. Consequently, the robot explores the domain almost uniformly at random, without adapting to the geometry of the objective function. Therefore, an appropriate choice of τ should enable both sufficient exploration and adaptation of the objective's geometry [40].

In our experiments, we selected two representative scenarios. The empty space represents open and unobstructed environments, while the structured space includes obstacles that further complicate the gas fields. Together, these scenarios span the typical range of environments in which the GSL method is expected to operate, providing guidance on how the temperature parameter τ can be chosen for practical applications.

We evaluated the impact of the temperature parameter in both scenarios and selected appropriate values accordingly. (a) Empty space. Temperature parameters for a team of robots should be chosen with appropriate magnitudes while maintaining sufficient diversity. To determine suitable magnitudes, we refer to Figure 13 in Section 7.3, where the impact of the temperature parameter is evaluated in an empty space. For experiments with three robots, we set their temperatures to $\{\tau_1, \tau_2, \tau_3\} = \{10^{-2}, 10^{-1}, 10^0\}$. (b) Structured space. Similarly, we evaluated different temperature settings and selected $\{\tau_1, \tau_2, \tau_3\} = \{10^{-1}, 10^0, 10^1\}$ for experiments with three robots. The larger magnitudes compared to scenario (a) can be attributed to the increased exploration required in a structured environment.

C Categorization of Stochastic Methods

We provide a categorization of Stochastic methods to show where our methods fits. Stochastic methods can be divided into five types below.

(i) *Sample average approximation (SAA)*. This is a model-based optimization method and requires explicit formulation of the cost function to solve an approximate optima. Specifically, SAA first samples a set of cost function realizations $\{f(x, \xi_i) | i = 1, 2, \dots, n\}$, where $f(\cdot)$ is the cost function. SAA then searches x that minimizes $\frac{1}{n} \sum_{i=1}^n f(x, \xi_i)$ as an approximation of the expected cost function. This method doesn't fit into our situation since our cost function (also known as potential function) is determined by the underlying patchy gas concentration field, which doesn't have an analytical model to describe its characteristics accurately. The most widely used analytical model

for air pollution dispersion, Gaussian Plume Model [24], is oversimplified and doesn't describe the patchiness.

(ii) *Stochastic approximation*, e.g., stochastic GD, are iterative methods used for root-finding problems [66]. Let g be the function of which we are finding the root, the decision updates recursively by $x_{i+1} = x_i - a_i g(x_i, \xi_i)$ where a_i is the moving distance. Stochastic GD is a specific instance of stochastic approximation, when g is the gradient of the cost function $\nabla_x f(x)$. Stochastic GD tends to get stuck in nonconvex settings, exhibiting a poor performance in multi-model landscapes, which is the case in patchiness-aware GSL.

(iii) *Global metamodel optimization*, e.g., **Bayesian optimization (BO)**. This approach utilizes global metamodels relating a set of inputs to an output through regressions [67]. An example is BO using Gaussian processes for regression [68]. Global metamodel optimization is done through a two-step process similar to SAA. N decision-response pairs are first collected to fit the metamodel, then a new decision is generated by minimizing the fitted metamodel. Like SAA, this method is not suitable for our problem due to complexity of the patchy gas distribution.

(iv) *Population-based optimization* maintains and evolves a population of candidate solutions over iterations. One of the most renowned methods is PSO, which is a heuristic algorithm inspired by the behavior of bird flock randomly searching for food [11]. Due to its simplicity and straightforward implementation, PSO has caught the attention of researchers as a way of solving various optimization problem and been widely applied in the field of swarm robotics. However, PSO suffers from limitations such as low exploration ability and lack of particle diversity, resulting in issues such as local minima and low-quality solutions. As stated in Section 2 of our manuscript, the shortage of PSO is that robots will have a tendency to move towards the same local optimum.

(v) *Sampling-based optimization* refers to algorithms that optimize an objective function by generating samples from some distribution. Given a Markov chain with equilibrium distribution $\pi(x)$, drawing samples from this distribution can be implemented by recording states from the chain (as stated in Section 5.1 of our manuscript). Monte Carlo sampling has been shown to be a powerful method for nonconvex and multimodal optimization [40]. Specifically, previous works have shown that there's a natural class of nonconvex problem for which the computational complexity scales more slowly than that of optimization algorithms [39]. Considering MC sampling methods' potential to handle nonconvex and multi-model objectives (an advantage over (ii) and (iv)), as well as its independency of an explicit analytic model for a patchy field (an advantage over (i) and (iii)), we build our strategy on this stochastic method. Moreover, replica exchange Langevin dynamics, a multiple Markov chain version extension to Langevin MC sampling, further provides a solution to multi-robot coordination, while backed by theoretical convergence analysis (an advantage over heuristic methods in (iv)).

Received 7 December 2024; revised 16 October 2025; accepted 23 November 2025